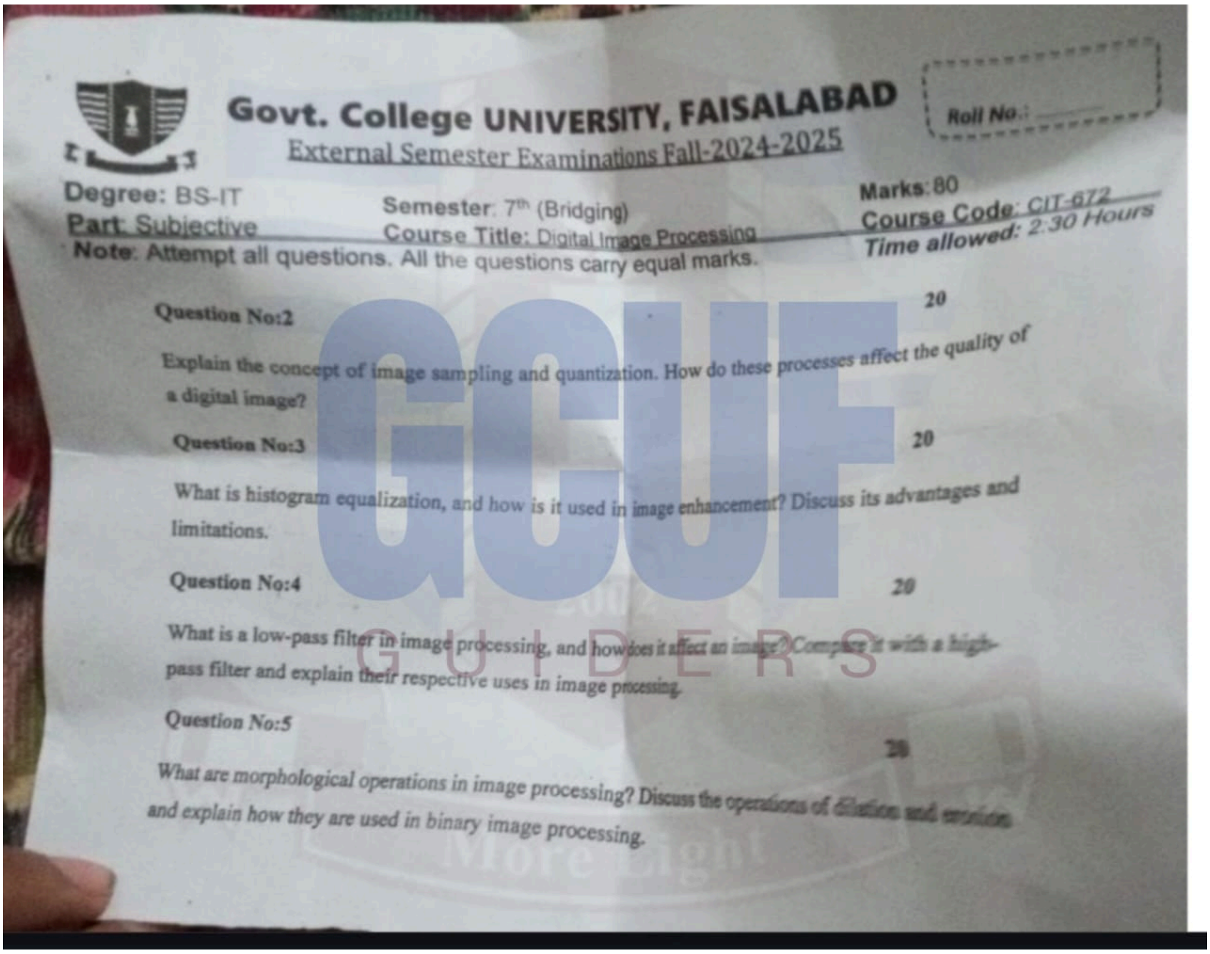

Govt. College UNIVERSITY, FAISALABAD

External Semester Examinations Fall-2024-2025

Roll No.:

Degree: BS-IT | Semester: 7th (Bridging) | Marks: 80

Part: Subjective | Course Title: Digital Image Processing | Course Code: CIT-672

Time allowed: 2:30 Hours

Note: Attempt all questions. All the questions carry equal marks.

Question No:2 — 20

Explain the concept of image sampling and quantization. How do these processes affect the quality of a digital image?

Question No:3 — 20

What is histogram equalization, and how is it used in image enhancement? Discuss its advantages and limitations.

Question No:4 — 20

What is a low-pass filter in image processing, and how does it affect an image? Compare it with a high-pass filter and explain their respective uses in image processing.

Question No:5 — 20

What are morphological operations in image processing? Discuss the operations of dilation and erosion and explain how they are used in binary image processing.

Question No. 2

Department: Information Technology
Course: Digital Image Processing
Marks: 20

Explain the concept of image sampling and quantization. How do these processes affect the quality of a digital image?

Introduction

Digital Image Processing deals with converting real-world continuous images into digital form so that they can be stored, processed, and transmitted by computers.

This conversion is achieved through two fundamental processes: **image sampling** and **image quantization**. These two steps determine the resolution, accuracy, and visual quality of a digital image.

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1. Image Sampling

Definition

Image sampling is the process of converting a continuous image into a discrete set of pixels by selecting a finite number of points along the spatial coordinates (x and y).**
In simple words, sampling decides **how many pixels** an image will contain.

Concept

A continuous image has infinite points. During sampling, the image is divided into a grid, and pixel values are taken at regular intervals:

$f(x,y) \rightarrow f(m,n)$

where
$m \text{ and } n$ are discrete spatial coordinates.

The number of samples taken per unit area is known as the **sampling rate**.

Effect on Image Quality

Sampling Level	Effect
High Sampling Rate	Produces high spatial resolution and sharp details
Low Sampling Rate	Causes loss of detail and produces distortion

Very Low Sampling Rate	Leads to **aliasing**, where edges appear jagged and false patterns occur

Aliasing is a major problem caused by insufficient sampling.

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2. Image Quantization

Definition

Quantization is the process of mapping continuous intensity values of pixels into a finite set of discrete gray levels.
It determines **how many intensity levels** a digital image can represent.

Concept

Pixel intensity values are approximated to the nearest allowed level:

$$L = 2^k$$

where
k = number of bits per pixel
L = number of gray levels

Examples:

Bits	Gray Levels
1-bit	2 levels (black & white)
4-bit	16 levels

8-bit	256 levels
16-bit	65,536 levels

Effect on Image Quality

Quantization Level	Effect
High Bit Depth	Smooth transitions, better contrast
Low Bit Depth	Produces **quantization error** and banding
Very Low Bit Depth	Causes false contours and detail loss

Quantization error occurs when pixel values are forced into limited gray levels.

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3. Combined Effect on Image Quality

Aspect	Sampling	Quantization
Controls	Spatial resolution	Gray level resolution

Poor Setting Causes	Blurring, aliasing	Banding, false contours
Improves	Image sharpness	Image smoothness and contrast

Both processes directly affect:

- Sharpness
- Detail clarity
- Noise appearance
- Storage requirements

Higher sampling and quantization increase image quality but also increase **memory usage and processing time**.

Conclusion

Image sampling and quantization are the two fundamental steps in converting an analog image into digital form. Sampling determines the number of pixels, while quantization determines the number of intensity levels. Improper sampling leads to aliasing and detail loss, while poor quantization results in banding and distortion. Therefore, choosing appropriate sampling rates and quantization levels is essential to achieve a high-quality digital image with efficient storage and processing.

Question No. 3

Department: Information Technology
Course: Digital Image Processing
Marks: 20

What is histogram equalization, and how is it used in image enhancement? Discuss its advantages and limitations.

Introduction

In Digital Image Processing, many images suffer from poor contrast due to improper illumination, shadows, or sensor limitations. **Histogram Equalization (HE)** is a widely used image enhancement technique that improves the contrast of an image by redistributing its intensity values. It makes important features more visible and enhances overall image quality.

Concept of Histogram Equalization

A **histogram** of an image represents the frequency distribution of pixel intensity values. Histogram equalization modifies this distribution so that the intensities are spread more uniformly across the available gray levels.

Definition

Histogram Equalization is a technique that transforms the gray-level values of an image so that its histogram becomes approximately uniform.

This leads to improved global contrast.

Working Principle

Let an image have intensity levels ranging from $0 \; to \; L - 1$.

1. Compute the histogram of the image.

2. Calculate the probability of each intensity level:

$$p(r_k) = \frac{n_k}{MN}$$

where
n_k = number of pixels with intensity r_k
$M \times N$ = image size

3. Compute the cumulative distribution function (CDF):

$$s_k = \sum_{j=0}^{k} p(r_j)$$

4. Map the original pixel values to new values:

$$s_k = (L - 1) \sum_{j=0}^{k} p(r_j)$$

This transformation spreads pixel values across the full intensity range.

Role in Image Enhancement

Histogram equalization improves:

• Visibility of details in dark or bright regions
• Overall contrast
• Perceptual quality
• Feature detection in medical, satellite, and industrial images

It is commonly used in:

- X-ray and MRI imaging
- Fingerprint enhancement
- Surveillance footage
- Low-light photography

Advantages

Advantage	Description
Improves contrast automatically	No need for manual parameter selection
Simple and fast	Computationally efficient
Enhances details	Makes hidden features visible
Effective for poorly illuminated images	Especially useful for low contrast images

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Limitations

Limitation	Description
Noise amplification	Enhances noise along with details
Over-enhancement	May produce unnatural appearance
Not suitable for all images	Fails when histogram is already uniform

Global operation	Cannot enhance local regions effectively
Loss of brightness information	Mean intensity may change

Conclusion

Histogram equalization is an important image enhancement technique that redistributes intensity values to achieve better contrast and visibility. It is simple and effective for low-contrast images but may amplify noise and produce unnatural results in some cases. Therefore, it must be used carefully, and in some applications, adaptive methods such as CLAHE are preferred.

Question No. 4

Department: Information Technology
Course: Digital Image Processing
Marks: 20

What is a low-pass filter in image processing, and how does it affect an image? Compare it with a high-pass filter and explain their respective uses in image processing.

Introduction

In Digital Image Processing, filtering is an important technique used to modify or enhance images by manipulating their frequency components. Two commonly used filters are **low-pass filters (LPF)** and **high-pass filters (HPF)**. These filters are used to control image smoothness, noise reduction, and edge enhancement.

Low-Pass Filter (LPF)

Definition

A **low-pass filter** is a filter that allows **low-frequency components** of an image to pass while attenuating or removing **high-frequency components**.

Low-frequency components represent slowly varying regions (smooth areas), while high-frequency components represent sharp transitions such as edges and noise.

Effect on an Image

LPF **smooths the image** by reducing noise and minor details.

Common Effects:

- Blurring and smoothing
- Reduction of random noise
- Loss of fine details and edges

Example Filters:

- Mean filter
- Gaussian filter
- Averaging filter

High-Pass Filter (HPF)

Definition

A **high-pass filter** allows **high-frequency components** to pass and suppresses low-frequency components.

High frequencies correspond to rapid intensity changes such as edges, lines, and fine details.

Effect on an Image

HPF **sharpens the image** by emphasizing edges and boundaries.

Common Effects:

- Enhances edges
- Highlights fine details
- Increases noise if not controlled

Example Filters:

- Laplacian filter
- Sobel filter
- Prewitt filter

Comparison between LPF and HPF

Feature	Low-Pass Filter	High-Pass Filter
Passes	Low frequencies	High frequencies
Removes	Noise and fine details	Smooth background

Effect	Blurring & smoothing	Sharpening & edge enhancement
Noise handling	Reduces noise	Amplifies noise
Output	Smooth image	Sharp and detailed image

Uses in Image Processing

Low-Pass Filter Uses	High-Pass Filter Uses
Noise reduction	Edge detection
Image smoothing	Image sharpening
Preprocessing for segmentation	Feature extraction
Blurring backgrounds	Medical and satellite image analysis

Conclusion

Low-pass and high-pass filters play opposite but complementary roles in image processing. Low-pass filters reduce noise and smooth images but may blur details, while high-pass filters enhance edges and details but may increase noise. The selection of the appropriate filter depends on the desired outcome in image enhancement and analysis.

Question No. 5

Department: Information Technology
Course: Digital Image Processing
Marks: 20

What are morphological operations in image processing? Discuss the operations of dilation and erosion and explain how they are used in binary image processing.

Introduction

Morphological operations are a class of non-linear image processing techniques that are based on the shape and structure of objects in an image. These operations are mainly applied to **binary images** and are widely used for object detection, noise removal, boundary extraction, and image segmentation. The two fundamental morphological operations are **dilation** and **erosion**.

Morphological Operations

Definition

Morphological operations are image processing techniques that modify the geometrical structure of objects in an image using a small predefined shape called a structuring element.

They focus on the spatial arrangement of pixels rather than their intensity values.

Structuring Element

A **structuring element (SE)** is a small binary matrix that defines how the image is probed and processed.
Common shapes include square, rectangle, cross, and circle.

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1. Dilation

Definition

Dilation is a morphological operation that expands or thickens the objects in a binary image. It increases the number of foreground (white) pixels.

Mathematical Representation

$$A \oplus B = \{z \mid (B)_z \cap A \neq \emptyset\}$$

Where:
A = input image
B = structuring element

Effect of Dilation

- Enlarges object boundaries
- Fills small holes and gaps
- Connects nearby objects
- Makes thin objects thicker

Applications

- Closing broken characters in text images
- Bridging gaps in edges
- Enhancing object connectivity
- Removing small black holes inside objects

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2. Erosion

Definition

Erosion is a morphological operation that shrinks or thins the objects in a binary image. It removes boundary pixels of objects.

Mathematical Representation

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

Effect of Erosion

- Reduces object size
- Removes small noisy white regions
- Separates connected objects
- Sharpens boundaries

Applications

- Eliminating small noise points
- Separating touching objects
- Removing thin protrusions
- Preprocessing before segmentation

Use in Binary Image Processing

Operation	Purpose
Dilation	Expands objects, fills holes
Erosion	Shrinks objects, removes noise
Erosion followed by Dilation (Opening)	Removes small objects
Dilation followed by Erosion (Closing)	Fills gaps and holes

Conclusion

Morphological operations are powerful tools in binary image processing that modify the shape and structure of objects using structuring elements. Dilation expands object regions while erosion shrinks them. Together, they are used for noise removal, object separation, feature extraction, and segmentation in digital image analysis.